CMAN:

Given data $N = 32$, $FP32 = 4$ bytes, DRAM bandwidth $= 320$ GB/s

Compute $= 10$ TFLOPS

Task 1:

$N = 32$ times, So each B[K][J] accessed 32 times

- A accesses : $N^3 = 32^3 = 32,768$ elements
- B accesses : $N^3 = 32^3 = 32,768$ elements

Total elements accessed :

$$2N^3 = 2 \times 32^3 = 65,536$$

Total DRAM bytes :

$$65,536 \times 4 = 262,144 \text{ bytes} = 256 \text{ KB}.$$

Total DRAM traffic (naive) $= 256$ KB.

Task 2:

Tile size $T = 8$

Number of tiles per dimension: $\frac{N}{T} = \frac{32}{8} = 4$

Each A and B tile is reused $N/T = 4$ times

A tile loads :

$$\left(\frac{N}{T}\right)^3 \times T^2 = (4)^3 \times 8^2 = 64 \times 64 = 4096 \text{ elements}$$

B tile loads :

4096 elements.

Total DRAM traffic:

$$2 \times 4096 \times 4 = 32{,}768 \text{ bytes} = 32KB$$

total DRAM traffic (tiled) : 32KB.

Task3:

Naive traffic = 262,144 bytes

Tiled traffic = 32,768 bytes

Traffic ratio: $\frac{\text{Naive}}{\text{Tiled}} = \frac{262{,}144}{32{,}768} = 8T$

If we consider only B's reuse:

$$\text{Tiled traffic} = \frac{262{,}144}{32} = 8{,}192 \text{ bytes}$$

Ratio: $\frac{262{,}144}{8{,}192} = 32 = N$

Explanation:

In the naive method, Matrix B is reloaded 32 times because each row of A requires reading all of B again.

Tiling keeps B in fast memory and reuses it, so B is loaded only once per tile. Therefore, naive traffic is exactly N times larger.

Task4:

Bandwidth = 320 GB/s

Compute = 10 TFLOPS.

Total FLOPs $= 2 \times N^3 = 2 \times 32^3 = 65,536$ FLOPs

Naive Case:

Memory time: $t_{memory} = \frac{\text{Traffic}}{\text{BW}}$

$$= \frac{2,62,144}{320 \times 10^9} = 8.192 \times 10^{-7} s = 0.819 \text{ ms}$$

Compute time:

$$t_{compute} = \frac{\text{FLOPs}}{P}$$

$$= \frac{65,536}{10^{13}} = 6.554 \times 10^{-9} s = 0.00655 \text{ ms}$$

t_{memory} (0.819ms) $\gg$ $t_{compute}$ (0.00655ms)

Execution time (naive) : 0.819ms

Bottleneck: Memory-bound.

Tiled Case (T=8)

Memory time:

$$t_{memory} = \frac{32,768}{320 \times 10^9} = 1.024 \times 10^{-7} s = 0.1024 \text{ ms}$$

Compute time: $\frac{\text{FLOPs}}{P} = \frac{65,536}{1.6 \times 10^{13}}$

$$\Rightarrow 6.554 \times 10^{-9} s = 0.00655 \text{ ms}$$

t_{memory} (0.1024 ms) > $t_{compute}$ (0.00655ms)

Execution time(tiled) = 0.1024 ms

Bottleneck : Memory-bound.